Pre-Stance Vibro-Compaction Improves Legged Locomotion on Granular Substrates

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***Abstract*—Legged robots lose mobility on granular substrates because foot loading can exceed the substrate yield force, causing sinkage and slip rather than useful propulsion. This project investigates whether a brief vibration phase, inserted immediately before the slow stance phase of a Buehler-clock gait, can compact the local granular bed and improve locomotion. We built a four-actuator quadruped with bent legs and wide flat feet, tested it in a shallow pit of 6 mm plastic BB pellets used as a surrogate for unstable granular sand, and measured performance with side-view computer vision. Across duration and frequency–amplitude sweeps, pre-stance vibration improved per-gait-cycle displacement and reduced slip relative to a no-vibration baseline. The best tested condition, 6 Hz at 3° servo amplitude with 1.4 s vibration duration, increased per-cycle displacement by 39.5% and reduced slip by 68% relative to baseline. These results support the hypothesis that phase-gated vibro-compaction can improve granular locomotion by increasing the effective local bearing capacity before stance, while also showing that vibration intensity cannot be summarized by a single scalar acceleration because the bent leg produces a spatial gradient of local vibration amplitudes.**

***Index Terms*—granular locomotion, vibro-compaction, legged robots, terradynamics, Buehler clock**

I. Introduction

Robots operating on deformable granular terrain often fail because the substrate yields under load. When the applied foot force exceeds the local yield force or bearing capacity of the granular bed, grains flow away from the contact patch, the robot sinks, and actuator work is dissipated into substrate rearrangement rather than forward motion [1], [2]. This solid–liquid duality is central to granular media: the same bed can resist load like a solid when force chains are stable, but flow like a liquid when those force chains are broken [3].

In this paper, we use yield force to mean the approximate load or shear-force threshold at which the granular contact patch stops behaving like a stable solid and begins to rearrange or flow. If the force applied by the robot foot remains below this threshold, grains can support the foot through force chains and the leg can push against the substrate. If the applied force exceeds this threshold, those force chains collapse, grains move around the foot, and the substrate behaves more like a fluid: the robot sinks, slips, and loses propulsive work to grain motion. Increasing local packing fraction through compaction should raise this effective threshold, making the same gait more likely to produce forward motion rather than substrate failure.

This project asks whether a legged robot can intentionally modify the local state of the substrate before pushing against it. Our central hypothesis is that pre-stance vibration causes local compaction, increasing the effective yield force of the granular material and therefore reducing slip during the subsequent propulsive stance phase. Rather than using vibration continuously for propulsion, we insert a brief vibro-compaction phase into the Buehler clock immediately before the slow stance phase. The vibration should coincide with first contact, compacting the grains beneath the foot before the robot asks that region of substrate to support shear.

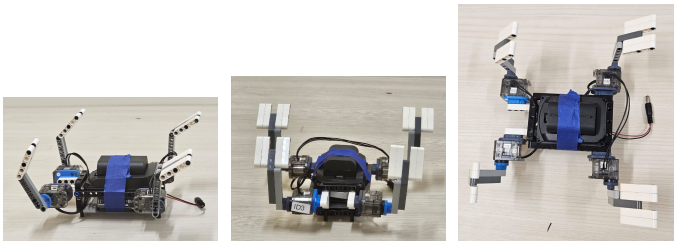
The experimental substrate is a shallow pit of 6 mm plastic BB pellets. This is not fine natural sand; it is a controlled, accessible surrogate for an unstable granular bed. Because BB pellets are comparatively large, heavy, and less compactable than finer sand, they likely produce a conservative test of the proposed mechanism. If measurable slip reduction appears in this coarse substrate, finer grains may show an equal or stronger response, though this remains future work.

The main contribution of this study is an experimental demonstration that phase-gated vibration can improve legged locomotion on a granular substrate. We show that the best tested vibration condition reduces slip by 68% and increases per-gait-cycle displacement by 39.5% relative to no-vibration walking. We also identify important limitations of a simple dimensionless acceleration model: because our leg rotates through an angle, the linear vibration amplitude varies from the joint to the toe, creating a "toe whip" effect and a gradient of local vibration regimes across the foot.

II. Related Work

Granular locomotion has been studied extensively through terradynamics and robophysics. Li *et al.* showed that leg geometry, intrusion depth, and substrate packing strongly determine reaction forces in granular media [2]. Qian *et al.* further connected appendage shape and gait design to locomotor performance on flowable ground [1]. These studies motivate our use of slip, displacement per gait cycle, and sinkage as mobility metrics.

Vibration is also well known to alter granular state. Civil engineering uses vibratory compaction to increase soil density and bearing capacity in foundations and road construction [4]–[6]. Granular physics studies show that vibration can produce either compaction or fluidization depending on vibration intensity and the resulting acceleration relative to gravity [7], [8]. As a useful rule of thumb, dimensionless acceleration below unity is often associated with compaction by friction breaking and settling, while stronger excitation can fluidize grains by overcoming gravitational confinement.



(a) Side (b) Front (c) Top

Fig. 1. Final robot morphology. The bent legs and long, wide feet increase contact area with the granular substrate. Red masts mounted to the chassis enable side-view computer-vision tracking.

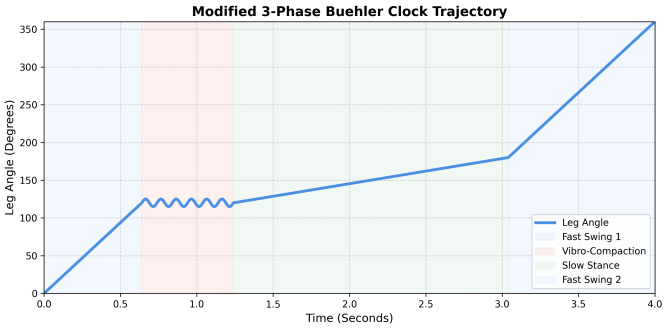


Fig. 2. Modified three-phase Buehler clock. A phase-gated vibration burst is inserted at contact before the slow stance phase, so the foot compacts the substrate before applying propulsive shear.

Robotics has used vibration in different ways. Vibrating bodies can self-propel through sand [9], ultrasonic vibration can reduce penetration resistance [10], and vibration can improve granular jamming gripper performance by helping grains settle into stronger configurations [11]. However, these uses are generally continuous or aimed at propulsion, penetration, or gripping. Our work instead applies vibration locally and briefly at the leg level, immediately before stance, to improve traction for a walking robot.

III. Robot and Gait Design

A. Robot Platform

The robot is a four-actuator quadruped built around Dynamixel XL-320 servos and Lego Technic structural elements. Each leg has a bent geometry with an interior knee angle of approximately 135°. The final morphology uses a 5.2 cm hip-to-ankle segment and a 4.0 cm long, relatively wide foot, producing an effective vibration lever arm of approximately 8.5 cm. The wide flat foot increases contact area and acts as a local compaction surface when vibration is applied at first contact.

Figure 1 shows the final morphology used in the experiments. The key features to note are the bent leg geometry, the long flat feet that increase contact area with the substrate, and the red masts used for side-view computer-vision tracking.

B. Modified Buehler Clock

The robot uses a Buehler walking gait generated by a phase clock with offsets across the four legs. The gait period is fixed at $T = 4.0\,\mathrm{s}$ with duty cycle $d_c = 0.45$, giving a slow propulsive stance duration of 1.8 s. The phase offsets are $\phi = [0, 0.25, 0.5, 0.75]$, producing a walking gait rather than a trot.

We modify the Buehler clock by inserting a vibration phase immediately before the slow propulsive phase:

1) **Fast swing:** the leg rapidly returns through the aerial portion of the cycle.
2) **Pre-stance vibration:** at contact, the macroscopic leg angle pauses for duration d while a sinusoidal angular vibration is superimposed on the servo command.
3) **Slow stance:** the leg pushes through the substrate at fixed slow angular velocity to generate forward motion.

This changes the clock trajectory, not the high-level walking gait. Vibration time is taken from the fast-swing portion of the cycle, so T and the slow-stance duration remain fixed. The slow stance angular velocity is therefore held constant at approximately $35\,^\circ\,\mathrm{s}^{-1}$, while the fast-swing velocity increases as d increases. This coupling is an important experimental limitation because longer vibration durations also demand faster swing return.

Figure 2 illustrates where the added vibration phase sits in the Buehler clock. The important point is that vibration occurs after fast swing and immediately before slow stance, so the same foot that compacts the substrate then pushes against it.

IV. Vibration Model and Toe-Whip Effect

The standard dimensionless acceleration for vertically vibrated granular media is

$$\Gamma = \frac{A(2\pi f)^2}{g}, \tag{1}$$

where A is linear vibration amplitude, f is vibration frequency, and g is gravitational acceleration. In idealized vertically shaken beds, $\Gamma < 1$ is often associated with compaction, while $\Gamma > 1$ can lead to fluidization [7], [8].

Our system only partially matches this idealization. The actuator command is angular, not linear. The local linear amplitude depends on distance from the servo axis:

$$A_v(R) \approx R\theta_{\mathrm{rad}}, \tag{2}$$

where R is the straight-line radius from the motor pivot to a point on the foot and θ_{rad} is the commanded angular amplitude in radians. The corresponding local commanded acceleration ratio is

$$\Gamma(R) = \frac{R\theta_{\mathrm{rad}}(2\pi f)^2}{g}. \tag{3}$$

For our 135° bent-foot morphology, the law of cosines gives approximate radii of $R_{\mathrm{heel}} = 0.052\,\mathrm{m}$, $R_{\mathrm{center}} = 0.0676\,\mathrm{m}$, and $R_{\mathrm{toe}} = 0.0851\,\mathrm{m}$. Thus, the same commanded vibration creates a spatial gradient: the heel may remain in a compacting regime while the toe can reach a locally fluidizing regime. We refer to this as a toe-whip effect. Because vibration

TABLE I
THEORETICAL COMMANDED Γ VALUES ACROSS THE 135° FOOT MORPHOLOGY. VALUES ARE CALCULATED FROM EQ. 3 USING THE HEEL, CENTER, AND TOE RADII.

f (Hz)	θ (deg)	Γ_{heel}	Γ_{center}	Γ_{toe}
6	2	0.263	0.342	0.430
6	3	0.394	0.513	0.646
6	4	0.526	0.684	0.861
8	2	0.467	0.608	0.765
8	3	0.701	0.912	1.148
8	4	0.935	1.215	1.530
10	2	0.730	0.950	1.195
10	3	1.096	1.424	1.793
10	4	1.461	1.899	2.391

stops before the slow stance phase, local toe fluidization is not necessarily catastrophic; it may disturb grains at the leading edge while the broader foot area still settles into a higher-bearing configuration before propulsion. This makes experimental parameter sweeps necessary rather than relying solely on Eq. 1.

Table I quantifies this gradient for the frequency–amplitude sweep. The reader should note that 6 Hz at 3°, the best measured condition, remains below the classical $\Gamma = 1$ threshold across the whole foot, while higher-frequency and higher-amplitude commands increasingly enter mixed or toe-fluidizing regimes.

V. EXPERIMENTAL METHODS

A. Testbed

Experiments were conducted in a shallow wooden pit filled with 6 mm plastic BB pellets. For the duration and frequency–amplitude sweeps, the pit was set to its base incline of approximately 2°. The substrate was raked level before every trial to reduce history effects. Each condition was tested in three trials unless otherwise noted due to visible gait glitches.

Figure 3 shows the physical testbed. The image is included to make clear that the BB pellets are a shallow surrogate granular substrate and that the camera observes the robot from the side rather than from above.

B. Computer-Vision Tracking

Side-view video was recorded at 120 fps using a fixed camera. Two red vertical masts mounted to the chassis were tracked frame-by-frame. The analysis pipeline slope-corrects the image, segments the red masts in HSV color space, identifies front and back mast centroids, and locks a pixels-per-meter calibration from the known mast separation of 18.2 cm. From these tracked points, the pipeline estimates horizontal displacement, pitch, front and back sinkage, instantaneous velocity, and per-cycle displacement.

Slip is computed from the ratio of actual forward velocity to a rigid-ground reference velocity:

$$s = 1 - \frac{v_{\text{actual}}}{v_{\text{rigid}}}, \tag{4}$$

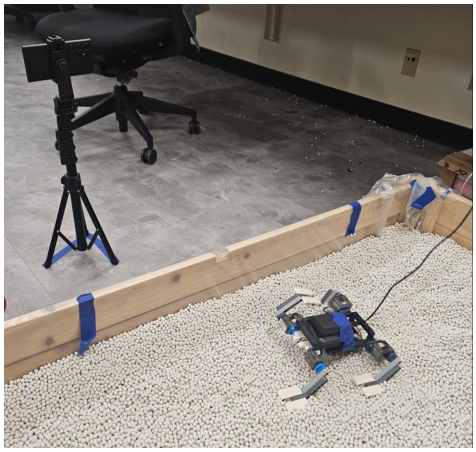


Fig. 3. The robot traverses a shallow pit of 6 mm plastic BB pellets while side-view video is recorded for mast-based tracking.



Fig. 4. Annotated computer-vision frame showing side-view mast detection, slope correction, pitch estimation, sinkage, and velocity overlay.

where $v_{\text{rigid}} = 0.048\,\text{m}\,\text{s}^{-1}$ was measured from no-vibration runs on rigid ground. After detecting the start of motion, the software segments each run into 4.0 s Buehler cycles and outputs cycle-level displacement, mean slip ratio, average velocity, sinkage, and pitch summaries. We use per-gait-cycle displacement and slip as the primary metrics. Sinkage and average velocity are retained as secondary diagnostics because they help interpret mechanism and failure modes.

Figure 4 shows a representative annotated frame from the tracking pipeline. The overlay highlights the mast detections, slope correction, pitch estimate, sinkage estimate, and instantaneous velocity used to generate the per-cycle outputs.

VI. RESULTS

A. Representative Diagnostic Traces

Before aggregating across trials, the computer-vision outputs were inspected to verify that the tracked motion corresponded to plausible locomotion and to identify physical signatures of improved mobility. Figure 5 shows a no-vibration baseline run. The front mast sinkage repeatedly reaches large negative values, approaching approximately −8 cm, indicating

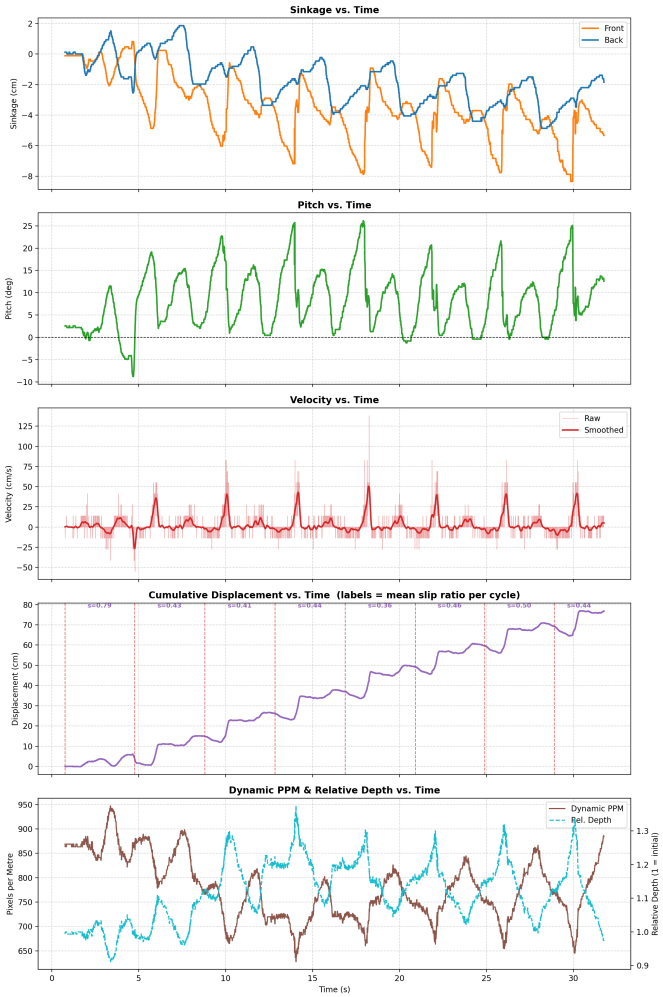


Fig. 5. Baseline diagnostic plot. The front mast repeatedly reaches large negative sinkage and the pitch trace shows large recurrent excursions, while per-cycle slip remains high throughout the run.

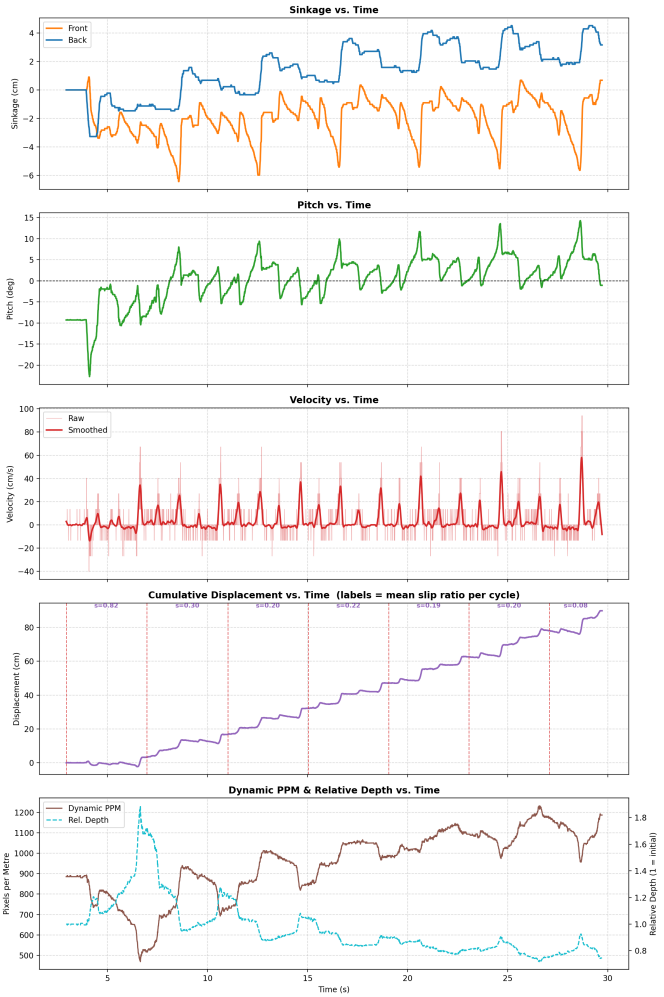


Fig. 6. Representative vibration diagnostic plot. The vibration run shows shallower absolute front sinkage, more bounded pitch after startup, and declining per-cycle slip, consistent with improved substrate support.

that the body pitches and settles substantially into the granular bed during stance. The pitch trace also shows recurrent large excursions, often reaching roughly 20° to 25°. Together, these diagnostics indicate that the no-vibration gait repeatedly overloads the substrate, causing both vertical yielding and body attitude variation. The displacement trace is divided into Buehler cycles, and the per-cycle slip labels remain high across the run, so repeated traversal alone does not produce a strong progressive improvement in the substrate.

Figure 6 shows a representative vibration trial at 10 Hz and $d = 1.4$ s. The diagnostic includes sinkage, pitch, velocity, cumulative displacement, and per-cycle slip. Compared with the baseline trace, the front mast generally exhibits smaller absolute sinkage, with the deepest excursions closer to −6 cm rather than −8 cm. The pitch trace is also more bounded after the initial startup transient, with fewer large positive excursions than the baseline run. These secondary diagnostics are consistent with the interpretation that pre-stance vibration increases the effective local bearing capacity of the substrate: the robot sinks less, rotates less severely, and converts more of each Buehler cycle into forward displacement. In this representative run, slip declines over repeated cycles, consistent with progressive substrate conditioning under repeated pre-stance vibration.

B. Experiment 1: Vibration Duration Sweep

The first experiment tested how long the robot should vibrate at contact before entering slow stance. Vibration duration was swept from 0.2 s to 1.4 s in 0.2 s increments at 6 Hz and 10 Hz, using constant servo amplitude $\theta = 2.5°$. A no-vibration Buehler-clock condition served as the baseline.

Figure 7 shows that per-gait-cycle displacement generally increases with vibration duration, with both frequencies reaching their best tested performance at $d = 1.4$ s. At this duration, 6 Hz achieves approximately 0.153 m cycle^{-1} and 10 Hz achieves approximately 0.148 m cycle^{-1}, compared with a baseline of approximately 0.122 m cycle^{-1}. Because the trend is still rising near the maximum tested duration, 1.4 s should

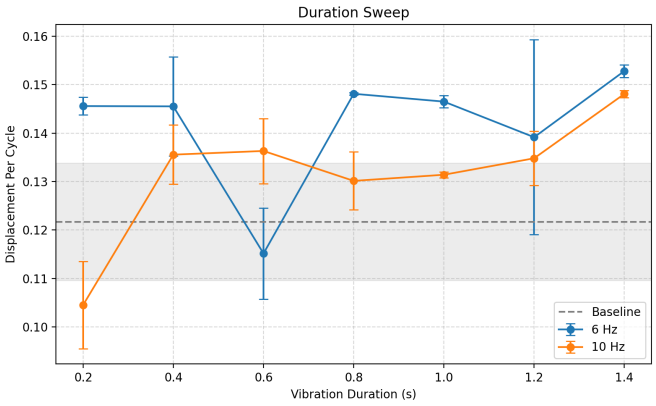


Fig. 7. Per-gait-cycle displacement versus vibration duration. Both tested frequencies perform best at the maximum tested duration, 1.4 s.

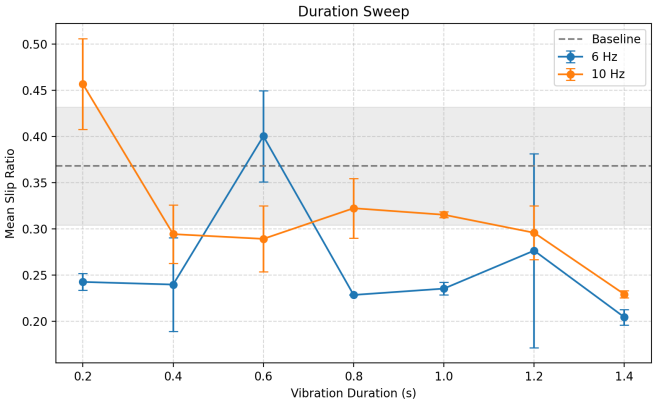


Fig. 8. Mean slip ratio versus vibration duration. Longer pre-stance vibration generally reduces slip, with the best tested values at 1.4 s.

be interpreted as the best tested duration rather than a proven global optimum.

The slip results in Fig. 8 show the same qualitative pattern. Most vibration conditions reduce slip relative to no vibration, with the best tested duration again at 1.4 s. The 6 Hz, 1.4 s condition reduces slip by approximately 44% relative to baseline. The 10 Hz, 0.2 s condition performs worse than baseline, suggesting that a very short burst at higher acceleration can disturb the foot placement before enough compaction occurs.

The duration sweep also suggests that the relevant compaction variable is closer to absolute vibration time than to the number of vibration oscillations. At the best tested duration, $d = 1.4$ s, the 6 Hz condition applies 8.4 vibration cycles while the 10 Hz condition applies 14 cycles. If performance were governed mainly by a fixed number of discrete taps, the two frequencies would be expected to peak at different durations but similar oscillation counts. Instead, both displacement and slip are best at the same duration. This suggests that the substrate response is better interpreted as a time-dependent relaxation or settling process under sustained agitation, rather than a purely cycle-count-based mechanism.

The main limitation of this sweep is actuator timing. The

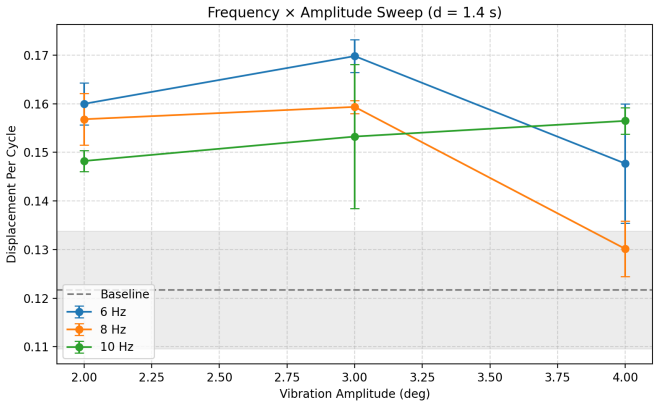


Fig. 9. Per-gait-cycle displacement for the frequency–amplitude sweep at $d = 1.4$ s. The best tested condition is 6 Hz at 3°.

Buehler period and slow-stance duration were held fixed, so increasing vibration duration shortened the available fast-swing interval. At $d = 1.4$ s, the required fast-swing velocity is already near the practical accuracy limit of the XL-320 servos. Testing longer durations would require either lengthening the gait period or reducing the swing angle.

C. Experiment 2: Frequency–Amplitude Sweep

The second experiment fixed $d = 1.4$ s and swept vibration frequency $f \in \{6, 8, 10\}$ Hz and servo amplitude $\theta \in \{2, 3, 4\}$ degrees. This tests the vibration parameter space near the best duration from Experiment 1. The corresponding theoretical commanded Γ values in Table I show that this sweep spans all-compacting, mixed, and toe-fluidizing regimes across the foot.

Figure 9 shows an inverted-U trend in amplitude. At 6 Hz, displacement rises from 2° to 3°, then falls at 4°. The best tested condition is 6 Hz at 3°, achieving approximately 0.170 m cycle^{-1}, a 39.5% improvement over baseline. The 8 Hz data show the same amplitude sweet spot, while the 10 Hz data are flatter and more variable.

Slip ratio, shown in Fig. 10, provides the clearest evidence for the vibro-compaction mechanism. Every tested vibration condition reduces slip relative to the baseline mean. The strongest reduction occurs at 6 Hz and 3°, where slip falls by approximately 68%. This condition is therefore selected as the best tested configuration because it simultaneously maximizes per-cycle displacement and minimizes slip.

Higher amplitude is not always better. Moving from 3° to 4° generally reduces performance, especially at 8 Hz. This is consistent with the toe-whip interpretation: at 8 Hz and 4°, the heel remains just below the classical $\Gamma = 1$ threshold while the center and toe exceed it. Larger angular amplitudes may therefore push part of the foot into a local fluidization regime even if other points along the leg remain compacting. Because vibration ends before slow stance, this local fluidization does not necessarily cause failure, but it can reduce the benefit of compaction.

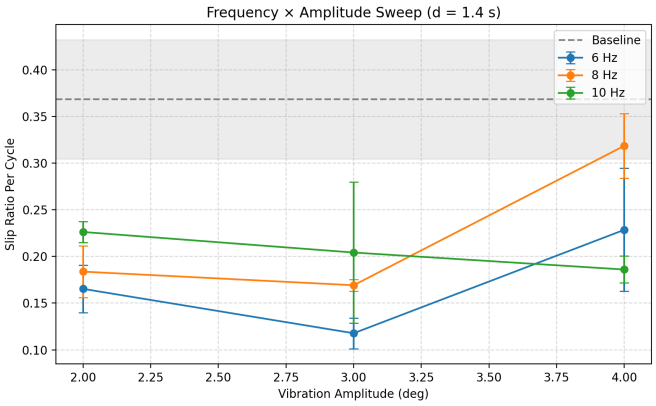


Fig. 10. Slip ratio for the frequency–amplitude sweep. The 6 Hz, 3° condition produces the largest slip reduction, approximately 68% relative to baseline.

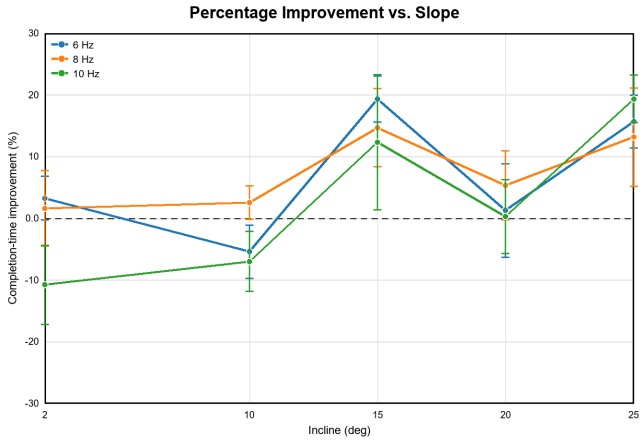


Fig. 11. Slope completion-time improvement with $\theta = 3°$ and $d = 1.4\,\mathrm{s}$, relative to the no-vibration baseline at the same slope.

D. Experiment 3: Slope Completion-Time Sweep

After identifying 3° as the best amplitude and 1.4 s as the best tested vibration duration, a slope sweep was performed to evaluate whether the effect persists as incline increases. In this experiment, amplitude was fixed at 3°, duration was fixed at 1.4 s, and frequencies of 6 Hz, 8 Hz, and 10 Hz were compared against a no-vibration baseline. Slopes of approximately 2°, 10°, 15°, 20°, and 25° were tested over a course length of approximately 1.2 m. Each condition used three trials, and all tested configurations completed the course.

Performance improvement was computed from completion time as

$$\text{Improvement} = \frac{t_{\text{baseline}} - t_{\text{vibration}}}{t_{\text{baseline}}} \times 100\%. \qquad (5)$$

Figure 11 shows the baseline-normalized completion-time improvement. The strongest and most consistent gains occur at the steeper inclines. At 15°, all vibration frequencies improve completion time: 6 Hz improves by 19.3%, 8 Hz by 14.7%, and 10 Hz by 12.3%. At 25°, all three frequencies again improve performance, with gains of 15.7%, 13.1%, and 19.3%, respectively. These results suggest that pre-stance vibration is most valuable when the baseline gait is traction-limited and the substrate is more likely to yield under load.

The lower-slope results are weaker and more mixed. At the base 2° incline, 6 Hz and 8 Hz provide small improvements of 3.2% and 1.6%, while 10 Hz is 10.8% slower than baseline. At 10°, 8 Hz gives a small 2.5% improvement, while 6 Hz and 10 Hz underperform baseline. The 20° results are also modest, with all three frequencies within approximately 5.3% of baseline. This non-monotonic behavior indicates that the optimal vibration parameters depend on slope-induced loading and body posture, not only on flat-pit slip reduction.

Importantly, this slope experiment was not a full four-dimensional optimization. The amplitude and duration were selected from the earlier sweeps performed at the base incline, then held fixed while slope and frequency varied. A complete sweep over slope, duration, amplitude, and frequency would have required far more trials than were feasible within the project timeline. The slope data should therefore be interpreted as a transfer test of the previously identified parameters: vibration generally helps under more difficult incline conditions, but slope-specific tuning would likely improve performance further.

VII. Discussion

A. Evidence for Pre-Stance Vibro-Compaction

The duration and frequency–amplitude sweeps support the hypothesis that pre-stance vibration improves granular locomotion by reducing slip. The mechanism is not simply that vibration makes the robot move faster; the slow stance phase is held fixed across conditions. Instead, vibration changes the substrate state before the foot applies propulsive shear. When the vibration parameters are appropriate, the substrate appears to provide higher effective bearing capacity, allowing more of the stance motion to become forward displacement. The slope sweep extends this result to a harder traversal task: vibration produces its clearest completion-time benefits at 15° and 25°, where baseline locomotion is slowest and traction limitations are most pronounced.

The best tested condition, 6 Hz at 3° for 1.4 s, lies near the upper edge of the all-compacting regime in Table I: the theoretical commanded values range from $\Gamma_{\text{heel}} = 0.394$ to $\Gamma_{\text{toe}} = 0.646$. This is physically plausible. Too little vibration may not break enough inter-particle friction to rearrange grains, while too much vibration may locally fluidize the toe region and weaken the contact. The optimal region is therefore expected to lie near a transition, where the bed is sufficiently agitated to settle but not so strongly disturbed that the stance substrate loses strength.

The same table also explains why Γ should not be treated as a complete predictor. The 8 Hz, 3° condition is mixed, with the toe slightly above unity, yet it still improves performance. The 10 Hz, 3° condition is above unity across the foot but still reduces slip relative to baseline. Several factors can explain this mismatch: the table uses commanded rather than measured

amplitude; XL-320 compliance, backlash, and load-dependent tracking error likely reduce actual vibration; Γ ignores granule size, mass, friction, and packing history; and the vibration stops before slow stance, giving locally disturbed grains time to resettle before propulsion. Thus, the gamma-gradient model supports the location of the best condition but does not replace experimental tuning.

B. Limitations

Several limitations remain. The testbed must be manually reset, and granular history may not be perfectly removed by raking. The BB pellets are large and less compactable than finer sand, making them an imperfect surrogate for natural granular terrain. The side-view camera provides useful pitch, displacement, and sinkage estimates, but onboard sensors or a depth camera would better capture three-dimensional body motion and foot-substrate interaction. The XL-320 servos also limit vibration resolution and accuracy, especially at higher amplitude, frequency, or duration.

VIII. Conclusion, Limitations, and Future Work

This project demonstrates that pre-stance vibro-compaction can improve legged locomotion on a granular substrate. By inserting a vibration phase into the Buehler clock immediately before slow stance, the robot reduced slip and increased per-cycle displacement relative to no-vibration walking. The best tested configuration was 6 Hz, 3°, and 1.4 s, which reduced slip by 68% and increased per-gait-cycle displacement by 39.5%.

Future work should test finer sand, use more precise motors, and study vibro-compaction with a static harness so yield-force changes can be measured directly without the confounds of full-body locomotion. A more complete multi-dimensional sweep over duration, frequency, amplitude, and slope would also test whether the 6 Hz, 3°, 1.4 s optimum generalizes across loading conditions.

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